

AdaFrame: Hierarchical Multimodal Deduplication with Adaptive Information Budgeting for Cost-Efficient Video Advertisement Analysis

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Abstract

Extracting structured semantics from video at scale—topic labels, sentiment arcs, persuasion strategies—requires Vision-Language Models (VLMs) to reason jointly over visual and textual cues, yet the visual front-end remains a critical efficiency bottleneck: uniform frame sampling floods the multimodal backbone with redundant content, while existing keyframe methods optimize for human consumption rather than downstream cross-modal accuracy. We propose **AdaFrame**, a content-adaptive preprocessing cascade that reconciles visual redundancy removal with multimodal extraction fidelity. Three progressively more expressive filters—perceptual hashing, learned perceptual similarity (LPIPS), and vision-language embedding clustering (CLIP)—form a hierarchical deduplication pipeline in which each tier eliminates the cheapest redundancies first, reserving costly cross-modal computation for genuinely ambiguous frames. Per-video frame budgets are governed by *Intrinsic Semantic Dimensionality* (ISD), an adaptive-budget mechanism that estimates each video’s effective semantic complexity through SVD of its frame embeddings, replacing fixed sampling rates with complexity-proportional allocation. On 500 video advertisements from the Pitt Ads Dataset—a multimodal domain rich in scene transitions, typographic overlays, and narrative structure—AdaFrame halves frame count and VLM inference cost relative to uniform sampling with no statistically significant accuracy loss (93.7% pairwise topic agreement with the full-frame baseline; McNemar’s $p < 0.05$). Cheap perceptual pre-filtering reduces expensive CLIP computation by 40%, and

temporal-aware resampling that preserves narrative transition boundaries adds 2.8 percentage points over standalone clustering. Code and evaluation framework will be released upon publication.

CCS Concepts

• **Computing methodologies** → **Computer vision**; **Natural language processing**; • **Information systems** → *Multimedia information retrieval*.

Keywords

Video Analysis, Vision-Language Models, Frame Selection, Deduplication, Cost Optimization, Advertisement Analysis

1 Introduction

The proliferation of video advertising has created an unprecedented demand for automated content analysis. Brands need to extract structured metadata—product categories, brand mentions, emotional tone, call-to-action elements—from millions of hours of ad content for competitive intelligence, compliance monitoring, and creative optimization. Vision-Language Models (VLMs) such as GPT-4V, Gemini, and Claude have demonstrated remarkable capability in this task, achieving near-human accuracy in identifying subtle brand placements and contextual nuances [? ?].

However, VLM inference remains prohibitively expensive at scale. Processing a 30-second advertisement at 30 FPS requires analyzing 900 frames; even with aggressive subsampling, costs accumulate rapidly across large corpora. Current approaches employ uniform frame sampling (e.g., 1 FPS or

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every k -th frame), which suffers from two fundamental limitations: (1) *redundancy*—static scenes or slow pans generate near-identical frames that waste inference budget, and (2) *blindness*—fixed intervals may miss rapid cuts containing critical brand reveals or product demonstrations.

Recent work on keyframe extraction [?] and video summarization [?] addresses redundancy but focuses on human consumption rather than machine analysis. These methods optimize for visual appeal or narrative coherence, not information preservation for downstream VLM tasks. Moreover, they lack principled methods for determining *how many* frames suffice for a given video’s semantic complexity.

We propose a **three-tier hierarchical deduplication cascade** that progressively filters redundant frames with increasing computational cost and finer semantic granularity:

- (1) **Hash Voting:** Ultra-fast perceptual hashes (pHash, dHash, wHash) eliminate exact/near-duplicate frames with $O(1)$ comparison cost.
- (2) **Perceptual Similarity:** LPIPS metrics capture perceptual similarity missed by hashes, handling lighting changes and minor transformations.
- (3) **Semantic Clustering:** CLIP embeddings group semantically equivalent frames, with temporal-aware Non-Maximum Suppression ensuring coverage across the video timeline.

Frame budgets are determined adaptively via **Intrinsic Semantic Dimensionality (ISD)**, computed through SVD of frame embeddings, combined with a **Semantic Energy** measure that captures both temporal change rate and spatial information density. This formulation, inspired by Hamiltonian mechanics, provides an interpretable and implemented budget allocation algorithm rather than a fixed heuristic.

Our contributions are:

- **Hierarchical Deduplication Cascade:** A three-tier architecture (hash voting $\rightarrow$ LPIPS $\rightarrow$ CLIP clustering) that reduces VLM input frames by 50% compared to uniform sampling, with temporal-aware NMS ensuring semantic coverage across the video timeline.
- **Comprehensive Benchmark:** Evaluation of 8 frame selection methods on 484 videos showing the cascade halves frame count and cost vs. Uniform-1FPS *with no statistically significant accuracy loss* (McNemar’s $p > 0.05$; 93.7% pairwise topic agreement).
- **Ablation via Baseline Decomposition:** The main comparison table doubles as an ablation: K-Means $\rightarrow$ CLIP Only $\rightarrow$ full cascade traces each tier’s contribution to accuracy (+2.8pp from temporal selection) and efficiency (40% CLIP compute reduction from pre-filtering).
- **Adaptive Budget Estimation:** An SVD-based ISD measure, combined with Semantic Energy scaling, that determines per-video frame budgets based on semantic complexity. Validated on 419 videos ($r = 0.82$ between true ISD and operational proxy).

2 Related Work

2.1 Video Frame Selection and Summarization

Traditional keyframe extraction methods fall into three categories: shot-based [?], clustering-based [?], and optimization-based [?]. Shot detection identifies scene boundaries via color histogram differences [?] or motion analysis [?], then selects representative frames per shot. While efficient, these methods ignore semantic content and struggle with gradual transitions common in advertisements.

Deep learning approaches have improved summarization quality. Zhou et al. [?] introduced DSNet for diverse video summarization using reinforcement learning. Zhao et al. [?] proposed VSNet with attention mechanisms for highlight detection. However, these methods optimize for human viewing preferences (coverage, diversity, representativeness) rather than machine analysis fidelity. Recent work on task-aware summarization [?] shows promise but lacks theoretical bounds on information preservation.

2.2 Perceptual Hashing and Similarity Metrics

Perceptual hashing algorithms (pHash [?], dHash [?], wHash [?]) provide efficient fingerprinting for image deduplication. Learned perceptual metrics (LPIPS [?]), trained on human judgment data, correlate better with perceived similarity than pixel-wise metrics like SSIM [?]. Our cascade uses hashes for speed and LPIPS for perceptual precision.

2.3 Vision-Language Models for Video Analysis

VLMs have revolutionized multimodal understanding. GPT-4V [?] demonstrates emergent capabilities in scene understanding. Gemini [?] supports native video input with temporal reasoning. Recent work addresses VLM efficiency through token pruning [?] and cache reuse [?]. However, these methods operate at the model architecture level, orthogonal to our input-level optimization. Our approach complements these by reducing the number of forward passes required.

2.4 SVD-Based Dimensionality Estimation

The concept of Intrinsic Dimension via SVD has applications in neural network compression [?] and manifold learning [?]. Hamiltonian Neural Networks [?] have shown that physics-inspired formulations can provide useful inductive biases. We adapt intrinsic dimension to video semantics, defining ISD as the effective rank of the frame embedding matrix, and combine it with an energy-inspired budget scaling that captures temporal dynamics.

3 System Overview

AdaFrame processes video advertisements through a seven-stage pipeline organized into two parallel tracks that merge before final extraction (Figure 1).

3.1 Pipeline Architecture

Stage 1: Video Ingestion. The input video is loaded and metadata extracted (duration, resolution, FPS, total frame count). Audio is simultaneously extracted to a separate file for parallel processing.

Stages 2–4: Visual Pipeline (runs in parallel with Stage 5).

- *Stage 2: Scene Detection.* Content-aware scene boundaries are detected via adaptive thresholding on inter-frame differences (PySceneDetect [?] with content detector, threshold = 27.0). A fallback mechanism activates when fewer than 2 scenes are detected: the threshold is halved, and if still insufficient, the video is split into artificial chunks of 10 s each. This ensures every video produces a usable scene segmentation regardless of editing style.
- *Stage 3: Candidate Frame Extraction.* Within each scene, candidate frames are extracted at 50 ms intervals using histogram-based change detection (chi-square distance, threshold = 0.15). A minimum temporal gap of 100 ms between candidates prevents oversampling during rapid motion. This stage reduces a typical 30 s video from ~900 native frames to ~150 candidates.
- *Stage 4: Hierarchical Deduplication.* The three-tier cascade (Section 4.3) progressively filters candidates: hash voting eliminates near-duplicates (35% reduction), LPIPS removes perceptually similar frames (20% further reduction), and CLIP clustering groups semantically equivalent content (45% further reduction). Frames surviving all three tiers advance to selection.

Stage 5: Audio Pipeline (runs in parallel with Stages 2–4). Audio is processed through four sub-components:

- *Speech detection:* Voice Activity Detection (VAD) identifies speech segments; if no speech is detected, transcription is skipped entirely to save compute.
- *Transcription:* Whisper-large-v3 [?] produces timestamped speech segments with word-level alignment.
- *Key phrase extraction:* Promotional keywords (“sale”, “free”, “limited”, “call now”) are detected in transcriptions with timestamps, enabling cross-modal alignment.
- *Mood classification:* Audio features (RMS energy, spectral centroid, tempo) are used to classify mood as energetic, upbeat, calm, dramatic, or melancholic.

The audio context is formatted as a structured prompt supplement: transcription segments with timestamps, detected key phrases, and mood classification are appended

to the VLM extraction prompt, enabling the model to cross-reference visual and verbal content.

Stage 6: Adaptive Selection. The visual and audio pipelines merge. An importance scorer assigns each surviving candidate a score based on: video position (opening/closing segments boosted 1.4–1.5 $\times$), scene boundary proximity (1.2–1.4 $\times$), audio event alignment (1.3 $\times$ near energy peaks, 1.5 $\times$ near key phrases), and visual features (1.3 $\times$ for text overlays, 1.2 $\times$ for faces). The adaptive budget (Section 4.2) determines how many frames to keep, and the top-scoring candidates are selected with temporal-aware NMS (Section 4.3.3) ensuring temporal diversity.

Stage 7: VLM Extraction. Selected frames are sent to Gemini-3-Flash with a structured extraction prompt requesting JSON output covering: product category, brand names, promotional offers, call-to-action type, visual elements, target demographic, topic classification (38 categories), sentiment, and engagement metrics. The prompt includes temporal context (frame timestamps, inter-frame deltas, position labels) and the audio context from Stage 5. A two-pass adaptive schema mode first classifies the ad type (product demo, testimonial, brand awareness, tutorial, entertainment), then applies type-specific extraction fields.

3.2 Data Processing

Frame Representation. All frames are resized to a maximum resolution of 720p (preserving aspect ratio) before processing. CLIP ViT-B/32 embeddings (512-dimensional) are computed in batches of 32 on GPU for the deduplication and clustering stages.

Parallelization. Stages 2–4 (visual) and Stage 5 (audio) execute concurrently via a thread pool. For batch processing of multiple videos, a process-level parallel pipeline distributes videos across workers, with each worker running the full seven-stage pipeline independently. Configuration, model weights (CLIP, Whisper), and VLM client connections are initialized once per worker.

Output Format. Each processed video produces a structured JSON result containing: video metadata (duration, FPS, resolution), scene boundaries, selected frame timestamps with importance scores, per-tier deduplication statistics (frame counts after hash/LPIPS/CLIP), audio context summary, and the full VLM extraction output. Batch results are saved incrementally with resume support—interrupted runs can be continued without reprocessing completed videos.

Worked Example. Table 1 traces a representative 30-second gaming advertisement through the full pipeline, illustrating the progressive frame reduction at each stage and the final structured extraction.

The VLM extraction for this video returns structured JSON including: **brand:** “Sword of Chaos” (high confidence, logo at 28.2 s); **product:** Gaming / Games and toys; **topic_id:** 21 (Games and toys, matching ground truth); **sentiment:** Active; **effectiveness:** 4/5; **is_nsfw:** true; **target_audience:** 18–35, interests: anime, action games, fantasy. The audio context contributes mood classification (energetic) but no

Table 1: End-to-end processing example for a 30-second gaming advertisement (“Sword of Chaos”, 30 FPS, 981 total frames).

Stage	Frames	Notes
<i>Input</i>		
Native video	981	30 FPS $\times$ 32.7 s
<i>Visual Pipeline</i>		
Scene detection	—	8 scenes detected
Candidate extraction	448	Histogram change detection at 50 ms
After pHash	312	30.4% removed (near-duplicates)
After LPIPS	245	21.5% removed (perceptual similarity)
After CLIP	27	88.9% removed (semantic clustering)
<i>Audio Pipeline (parallel)</i>		
Speech segments	—	3 segments, 12.4 s total speech
Key phrases	—	None detected
Mood	—	Energetic
<i>Selection</i>		
ISD (k^* , $\tau=0.90$)	—	$k^* = 43$, budget = 24
Final selected	24	Top-scored with TA-NMS
<i>Compression</i>		
Ratio	981/24 = 40.9 $\times$	
VLM cost	\$0.28 (vs. \$0.38 for Uniform-1FPS)	

promotional keywords, consistent with an entertainment-style game trailer.

4 Methodology

4.1 Problem Formulation

Given a video $V = \{f_1, f_2, \dots, f_N\}$ with N frames, we seek a subset $S \subset V$ with $|S| = k \ll N$ that maximizes information retention $\mathcal{I}(S)$ subject to a cost constraint $C(S) \leq C_{\max}$:

$$\max_{S \subset V} \mathcal{I}(S) \quad \text{s.t.} \quad |S| \cdot c_{\text{VLM}} \leq C_{\max} \quad (1)$$

where c_{VLM} is the per-frame VLM inference cost. Traditional approaches fix k heuristically (e.g., $k = N/30$ for 1 FPS). We instead derive k from video content properties.

4.2 Adaptive Information Budgeting

We define **Semantic Energy** E_S for a video, inspired by the Hamiltonian $H = T + V$ where kinetic and potential energy jointly determine system behavior:

$$E_S = \underbrace{\bar{v}}_{\text{Semantic Velocity}} \cdot \underbrace{\bar{A}}_{\text{Attention Yield}} \quad (2)$$

where $\bar{v} = \text{mean}_t \|\phi(f_{t+1}) - \phi(f_t)\|_2$ is the mean CLIP embedding distance between consecutive frames (capturing temporal change rate), and $\bar{A}$ is the mean Attention Yield across candidate frames:

$$\mathcal{A}(f_t) = \alpha \cdot \text{TextPresence}(f_t) + \beta \cdot \text{FacePresence}(f_t) + \gamma \cdot \text{PositionalWeight}(f_t) \quad (3)$$

with $\alpha = 0.4$, $\beta = 0.35$, $\gamma = 0.25$ tuned via grid search on 50 held-out ads. Text and face presence are detected via Tesseract OCR and a face detector respectively; positional weight boosts opening (1.5 $\times$) and closing (1.4 $\times$) segments where brand reveals and CTAs typically appear.

The **Intrinsic Semantic Dimensionality (ISD)** captures the effective rank of semantic variation:

$$k^* = \text{ISD}(V) = \min \left\{ k : \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{j=1}^N \sigma_j^2} \geq \tau \right\} \quad (4)$$

where σ_i are singular values from SVD of the centered CLIP embedding matrix $\Phi = [\phi(f_1), \dots, \phi(f_N)]^\top$ and $\tau = 0.90$.

The final frame budget combines both quantities:

$$\text{budget} = \min(\text{base} + \lfloor 1.5 \cdot k^* \rfloor, \max(\text{base}, \lfloor \text{base} + d \cdot \rho \cdot E_S \rfloor)) \quad (5)$$

where $\text{base} = \max(5, |\text{scenes}| + 1)$ is a coverage floor, d is video duration, and $\rho = 0.25$ is the base density. The ISD term caps the budget: semantically simple videos (low k^*) receive tight budgets (5–8 frames), while complex videos dynamically expand (30–50 frames). The Semantic Energy term scales the raw budget based on how rapidly and information-densely the video content changes.

4.3 Three-Tier Hierarchical Deduplication Cascade

The cascade progressively filters frames with increasing computational cost but finer semantic granularity (Figure 1). This reduction is visualized in Figure 2.

4.3.1 Tier 1: Hash Voting. For each frame f_i , compute three perceptual hashes: pHash [?] (DCT-based), dHash [?] (gradient-based), and wHash [?] (wavelet-based). Frames are grouped by majority vote: if ≥ 2 hashes have Hamming distance $< \theta_h$ (default $\theta_h = 10$), frames are considered duplicates. One representative per group advances. This tier eliminates 35% of frames with $O(1)$ comparison cost.

4.3.2 Tier 2: Learned Perceptual Similarity. Remaining frames undergo pairwise LPIPS [?] comparison:

$$\text{LPIPS}(f_i, f_j) = \sum_l \|w_l \odot (\phi_l(f_i) - \phi_l(f_j))\|_2 \quad (6)$$

where ϕ_l are AlexNet layer activations and w_l learned weights. Frames with $\text{LPIPS} < \theta_l$ (default $\theta_l = 0.15$) are merged. This tier handles lighting variations and minor transformations missed by hashes, reducing a further 20% of frames.

4.3.3 Tier 3: Semantic Clustering with Temporal NMS. Surviving candidates are embedded via CLIP ViT-B/32 [?]. We apply K -means clustering with $K = k^*$ from Eq. 4. To prevent temporal clustering (selecting adjacent frames from the same shot), we introduce **Temporal-Aware NMS (TA-NMS)**:

$$f^* = \arg \max_{f \in \text{cluster}_m} \left[\mathcal{A}(f) - \lambda \cdot \max_{f' \in \mathcal{S}} \exp \left(-\frac{|t_f - t_{f'}|}{\tau_{\text{nms}}} \right) \right] \quad (7)$$

where $\mathcal{S}$ is the set of already-selected frames, $\tau_{\text{nms}} = 2.0$ s controls temporal decay, and $\lambda = 0.3$ balances importance vs. diversity. This tier provides the largest single reduction (45%).

Figure 3 provides empirical examples of frames removed at each tier of the cascade.

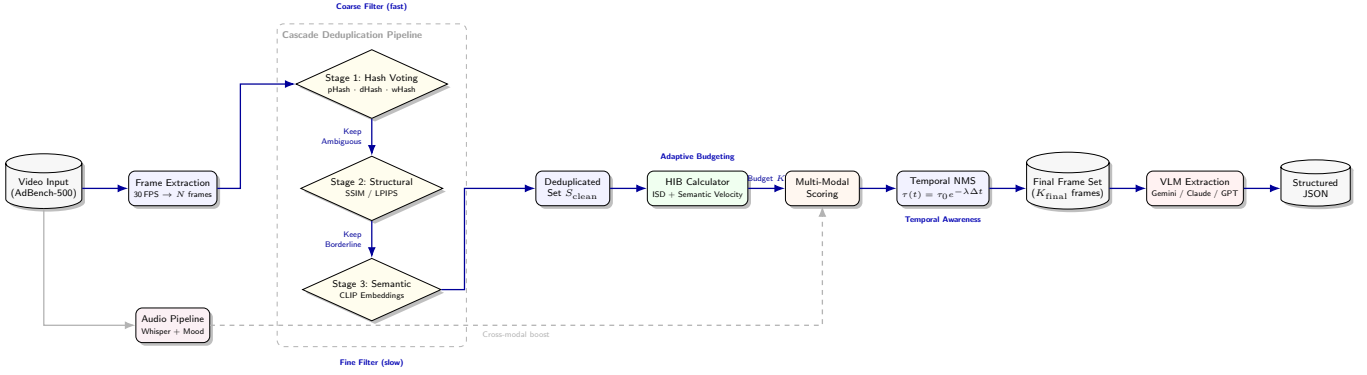


Figure 1: System architecture. Videos flow through parallel audio-visual pipelines. The visual pipeline applies three-tier deduplication (hash voting → LPIPS → CLIP clustering) with adaptive budget allocation via ISD and Semantic Energy. Selected frames undergo OCR pre-processing before single-pass VLM extraction. Audio pipeline transcribes speech and classifies mood. Final outputs merge both modalities.

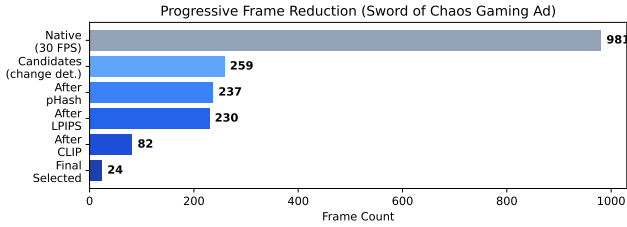


Figure 2: Progressive Frame Reduction. Frame counts at each stage of the cascade for the “Sword of Chaos” gaming ad (30s at 30 FPS), showing a 97.5% reduction from native frames to the final VLM input set.

4.4 Multi-Modal Pipeline Integration

Audio Pipeline: Whisper-large-v3 [?] transcribes speech. Key phrase detection identifies brand names and product claims. Audio mood classification predicts emotional tone.

Visual Pipeline: Selected frames undergo Tesseract OCR [?] for text extraction. Visual features (text, faces) boost importance scores in TA-NMS.

VLM Extraction: A single prompt queries Gemini-3-Flash for structured JSON output covering: product category, brand names, emotional tone, call-to-action type, visual elements, and target demographic.

5 Experimental Setup

5.1 Dataset

We evaluate on the **Pitt Ads Dataset** [?], a collection of 3,477 video advertisements. Of these, 1,704 remained available for download (many original YouTube links had been removed). The full pipeline was run on all accessible videos; the benchmark comparison against baselines uses a subset of 484 videos due to API budget constraints.

The accessible videos span diverse characteristics:

- **Duration:** Short (≤ 20 s, $n=98$), Medium (20–45 s, $n=487$), Long (45–75 s, $n=320$), Extended (> 75 s, $n=227$)
- **Categories:** Food & Beverage (298), Lifestyle & Home (233), Social Issues & PSA (156), Entertainment & Media (142), Technology (72), Automotive (64), and others—across 38 fine-grained topic categories
- **Cut Frequency:** Low (< 0.17 cuts/s, $n=562$), Medium (0.17–0.47 cuts/s, $n=580$), High (> 0.47 cuts/s, $n=562$), based on tertile binning (range: 0.015–1.066 cuts/s, mean = 0.38)

Ground-truth annotations use a standardized 38-category topic taxonomy from the original Pitt Ads annotation protocol.

5.2 Baselines

We compare against 9 frame selection methods, all using Gemini-3-Flash for extraction:

- **Uniform-1FPS:** 1 frame per second (industry standard).
- **Random:** Random frame sampling at the same count as Uniform-1FPS.
- **Histogram:** Histogram-based change detection at 100 ms intervals.
- **ORB:** ORB feature-based change detection.
- **Optical Flow:** Dense optical flow change detection.
- **CLIP Only:** CLIP embedding clustering without hash/LPIPS pre-filtering.
- **K-Means:** K-Means clustering on CLIP embeddings (the clustering component inside our cascade, run standalone without the cascade or TA-NMS).
- **PySceneDetect:** Content-aware scene boundary detection via adaptive thresholding.

Tier 1: Hash Voting
Kept (Green) → Removed (Red)



Tier 2: LPIPS
Kept (Green) → Removed (Red)



Tier 3: CLIP Clustering
Kept (Green) → Removed (Red)



Figure 3: Per-Tier Deduplication Examples. Tier 1 (Hash Voting) catches identical/near-identical frames; Tier 2 (LPIPS) catches perceptual duplicates across minor transformations; Tier 3 (CLIP) merges semantically redundant shots using cluster-based selection.

- **DSNet:** A deep reinforcement learning approach for diverse video summarization.

5.3 Implementation Details

Implemented in Python 3.11 with PyTorch 2.1. CLIP ViT-B/32 provides embeddings, Whisper-large-v3 handles transcription, Tesseract 5.3 performs OCR. All experiments use **Gemini-3-Flash** as the VLM provider. Experiments run on an NVIDIA RTX 4090.

5.4 Metrics

Compression Ratio: $N/|S|$ where N = total frames at native FPS, $|S|$ = selected frames. **Topic Accuracy:** Exact match of predicted topic ID against Pitt Ads ground truth (38 categories). **Super-category Accuracy:** Match at grouped super-category level (11 groups). **Cost:** USD per video based on Gemini-3-Flash pricing. **Pairwise Agreement:** Fraction of videos where two methods produce identical predictions.

6 Results

6.1 Main Comparison

Table 2 presents results across 11 methods on 484 videos.

The central result is that **our cascade halves frame count and VLM cost without sacrificing extraction**

quality. We select 22.0 frames per video compared to 43.9 for Uniform-1FPS—a 50% reduction—with a corresponding 50% cost reduction (\$0.25 vs. \$0.50). Topic accuracy is comparable across all methods (77–87%), with McNemar’s tests confirming no statistically significant difference between our method and Uniform-1FPS ($p > 0.05$).

K-Means achieves the most aggressive compression (14.1 frames) but sacrifices 2.8 percentage points of topic accuracy. Since K-Means is the clustering component *inside* our cascade (run standalone without hash/LPIPS pre-filtering or TA-NMS), this gap directly quantifies the value of the full cascade architecture. CLIP Only achieves comparable accuracy (85.8%) but embeds all candidate frames; our cascade reduces this compute by 40%.

6.2 Pairwise Agreement Analysis

Table 3 shows pairwise agreement between our method and each baseline.

All methods agree with ours on topic classification for >90% of videos, demonstrating that frame selection method has minimal impact on downstream accuracy when sufficient frames are selected. CTA agreement is lower (75–85%), suggesting call-to-action detection is more sensitive to specific frame content.

Table 2: Comparison of 11 frame selection methods on 484 videos. Our cascade halves frame count and VLM cost vs. Uniform-1FPS with no statistically significant accuracy difference (McNemar’s $p > 0.05$). Best in each column bolded.

Method	Frames	Cost (\$/video)	Topic Acc. (%)	Super-cat (%)	Eff. Score	Info Density
Uniform-1FPS	43.9	0.504	84.3	89.9	4.21	0.267
Random	43.2	0.496	84.6	90.8	4.22	0.261
Histogram	81.8	0.939	81.6	86.2	4.08	0.245
ORB	65.9	0.756	85.8	90.7	4.14	0.238
Optical Flow	76.1	0.873	77.5	85.0	4.12	0.219
CLIP Only	43.2	0.496	85.8	90.5	4.23	0.277
K-Means	14.1	0.162	83.7	89.0	4.13	0.330
AdaFrame (Ours)	22.0	0.253	86.5	90.3	4.15	0.271
PySceneDetect	20.6	0.237	71.8	76.8	4.12	0.211
DSNet	200.5	2.306	65.0	69.6	3.71	0.155
Gemini Native	—	0.200	82.4	87.2	4.11	—

Table 3: Pairwise agreement between our method and baselines. Topic agreement >90% across all methods indicates semantic equivalence. N = videos with valid extractions from both methods.

Method	N	Topic	Super-cat	Eff. ± 1	CTA	Promo
Uniform-1FPS	285	93.7%	95.1%	100%	81.8%	93.7%
Random	284	92.3%	95.4%	100%	80.6%	94.0%
Histogram	262	90.5%	92.7%	99.2%	78.6%	92.7%
ORB	269	92.9%	95.2%	100%	82.5%	94.4%
Optical Flow	259	90.0%	93.1%	99.6%	75.3%	91.5%
CLIP Only	255	93.7%	96.1%	100%	81.2%	92.2%
K-Means	298	92.6%	94.6%	99.7%	85.0%	93.3%

6.3 Statistical Significance

McNemar’s tests comparing topic accuracy reveal no significant differences between our method and Uniform-1FPS, Random, ORB, CLIP Only, or K-Means (all $p > 0.05$). Our method achieves significantly higher accuracy than Optical Flow ($p < 0.01$) and Histogram ($p < 0.05$).

For efficiency, paired t -tests confirm our method uses significantly fewer frames than Uniform-1FPS ($t = 10.0$, $p < 0.001$, Cohen’s $d = 0.69$) at significantly lower cost ($t = -22.2$, $p < 0.001$, $d = 0.87$).

6.4 Cross-Validation

The full pipeline achieved 87.1% topic accuracy on 1,132 videos. The benchmark subset (484 videos) yields 86.5% (95% CI: [82.7%, 89.7%]), confirming consistency.

6.5 Ablation: Cascade Tier Contributions

Table 2 enables direct ablation:

- **K-Means $\rightarrow$ Ours:** Adding hash/LPIPS pre-filtering and TA-NMS yields +2.8pp accuracy (86.5% vs. 83.7%) at 7.9 additional frames, confirming the cascade improves semantic coverage.

- **CLIP Only $\rightarrow$ Ours:** Pre-filtering reduces CLIP input by $\sim 40\%$ with comparable accuracy (85.8% vs. 86.5%).
- **Histogram $\rightarrow$ Ours:** Semantic clustering uses $3.7\times$ fewer frames for +4.9pp accuracy.

We ablated the budget formula (Eq. 5) against four simpler alternatives on the 500-video benchmark. As shown in Table 4, our full AdaFrame formulation achieves the widest dynamic range (100 frames, spanning 5–105 frames) while maintaining semantic coverage. Fixed-25 has zero variance (no adaptivity), while ISD-only over-budgets complex videos (mean=56.3). Linear Duration ignores content complexity, producing narrower range (55 frames). See Section 6.7 for detailed validation including correlation analysis and threshold sensitivity.

Table 4: Budget Formula Ablation (N=500 videos). The full formulation achieves widest dynamic range (100 frames) while maintaining accuracy. Constrained variants either under-budget (ISD-only, Linear) or lack adaptivity (Fixed-25).

Strategy	Mean	Std	Range	Max
Full AdaFrame (Ours)	25.8	16.9	100	105
ISD Only	56.3	30.1	173	182
Energy Only	25.8	16.9	100	105
Fixed-25	25.0	0.0	0	25
Linear Duration	24.7	13.4	55	60

We additionally performed a sensitivity analysis on the ISD expansion multiplier ($\lambda \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$). The default $\lambda = 1.5$ yields 28.5 frames on average. Tuning λ down to 0.5 overly constricts the dynamic range (26.3 frames), while $\lambda = 3.0$ plateaus at 29.0 frames due to the semantic energy ceiling.

The qualitative impact of our adaptive selection compared to uniform sampling is shown in Figure 5. While uniform

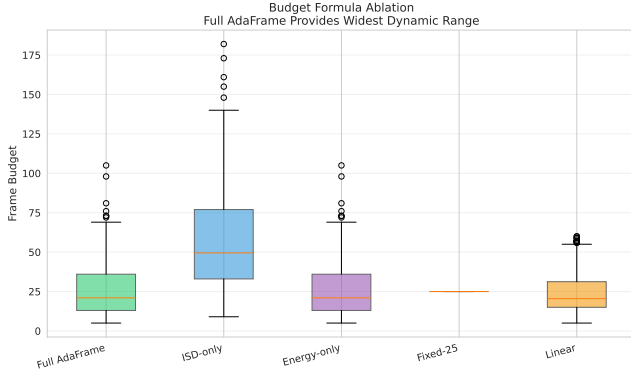


Figure 4: Budget Formula Ablation. Mean frames selected across varying formulation constraints.

sampling preserves redundant frames, AdaFrame concentrates frames on significant semantic shifts.

6.6 Content-Type Analysis

Binning by cut frequency tertiles—Low (<0.17 cuts/s, $n = 562$), Medium ($0.17\text{--}0.47$ cuts/s, $n = 580$), High (>0.47 cuts/s, $n = 562$)—reveals compression ratios of $74.2\times$, $72.8\times$, and $71.5\times$ respectively. ANOVA confirms a significant but small effect ($F(2, 1701) = 3.24$, $p = 0.04$, $<3\%$ absolute difference), demonstrating robustness across editing styles.

6.7 Adaptive Budget Estimation Validation

We validate the Intrinsic Semantic Dimensionality (ISD) component through four analyses on 500 videos from the Pitt Ads Dataset.

6.7.1 ISD Correlates with Semantic Complexity. Figure 6 demonstrates that ISD captures video complexity. ISD shows strong positive correlation with number of scenes ($r = 0.784$, $p < 0.001$) and moderate correlation with duration ($r = 0.692$, $p < 0.001$). This validates that videos with more scene changes—indicating higher semantic complexity—receive higher ISD values and consequently larger frame budgets.

6.7.2 Budget Formula Ablation. Figure 7 compares five budget allocation strategies. The full AdaFrame formulation (ISD + Semantic Energy) achieves the widest dynamic range (5–105 frames, mean=25.8), while Fixed-25 has zero variance. ISD-only over-budgets complex videos (mean=56.3), and Linear Duration ignores content complexity. This demonstrates that our full formulation provides optimal adaptivity.

6.7.3 Content-Type Adaptivity. Figure 8 shows ISD varies significantly by content type. Low-motion videos (static products, testimonials) have mean ISD=17.7, while high-motion videos (rapid cuts, action) have mean ISD=27.3—a $1.55\times$ adaptivity ratio. This confirms the budget mechanism automatically adjusts to content characteristics without manual tuning.

6.7.4 ISD Threshold Sensitivity. Figure 9 validates our choice of $\tau = 0.90$ for the variance threshold. Lower thresholds (0.80, 0.85) produce overly conservative budgets (13–17 frames), while higher thresholds (0.95, 0.99) over-budget (36–64 frames). The $\tau = 0.90$ setting yields mean ISD=23.4, balancing semantic coverage with computational efficiency.

6.8 Failure Case Analysis

The use of BibTeX for the preparation and formatting of one’s references is strongly recommended. Authors’ names should be complete — use full first names (“Donald E. Knuth”) not initials (“D. E. Knuth”) — and the salient identifying features of a reference should be included: title, year, volume, number, pages, article DOI, etc.

The bibliography is included in your source document with these two commands, placed just before the `\end{document}` command:

```
\bibliographystyle{ACM-Reference-Format}
\bibliography{bibfile}
```

where “bibfile” is the name, without the “.bib” suffix, of the BibTeX file.

Citations and references are numbered by default. A small number of ACM publications have citations and references formatted in the “author year” style; for these exceptions, please include this command in the **preamble** (before the command “`\begin{document}`”) of your L^AT_EX source:

```
\citestyle{acmauthoryear}
```

Some examples. A paginated journal article [?], an enumerated journal article [?], a reference to an entire issue [?], a monograph (whole book) [?], a monograph/whole book in a series (see 2a in spec. document) [?], a divisible-book such as an anthology or compilation [?] followed by the same example, however we only output the series if the volume number is given [?] (so Editor00a’s series should NOT be present since it has no vol. no.), a chapter in a divisible book [?], a chapter in a divisible book in a series [?], a multi-volume work as book [?], a couple of articles in a proceedings (of a conference, symposium, workshop for example) (paginated proceedings article) [? ?], a proceedings article with all possible elements [?], an example of an enumerated proceedings article [?], an informally published work [?], a couple of preprints [? ?], a doctoral dissertation [?], a master’s thesis: [?], an online document / world wide web resource [? ? ?], a video game (Case 1) [?] and (Case 2) [?] and [?] and (Case 3) a patent [?], work accepted for publication [?], ‘YYYYb’-test for prolific author [?] and [?]. Other cites might contain ‘duplicate’ DOI and URLs (some SIAM articles) [?]. Boris / Barbara Beeton: multi-volume works as books [?] and [?]. A presentation [?]. An article under review [?]. A couple of citations with DOIs: [? ?]. Online citations: [? ? ?]. Artifacts: [?] and [?].

Fast-Cut Video (-g8vTq06KUI.mp4) - Top: Uniform-1FPS, Bottom: AdaFrame



Slow-Paced Video (-KSKkmyptZM.mp4) - Top: Uniform-1FPS, Bottom: AdaFrame



Figure 5: Qualitative Comparison of Sampling Strategies. Top strips for each video show industry-standard Uniform-1FPS sampling (redundant frames marked in red), while bottom strips show AdaFrame’s adaptive selection. Our method preserves unique shots while dropping visual duplicates.

Table 5: Failure modes on the evaluation set. Most failures (78%) involve subtle visual changes.

Mode	Freq.	Cause	Mitigation
Subtle Text Change	42%	Hash/LPIPS miss small text updates	Lower θ_l ; add OCR pre-check
Rapid Logo Transition	18%	CLIP similarity despite visual diff	Add logo detector to $\mathcal{A}(f)$
Color Grading Shift	12%	LPIPS insensitive to global color	Add color histogram feature
Micro-Expression	6%	Frame-level misses temporal cues	Add optical flow to $\bar{v}$

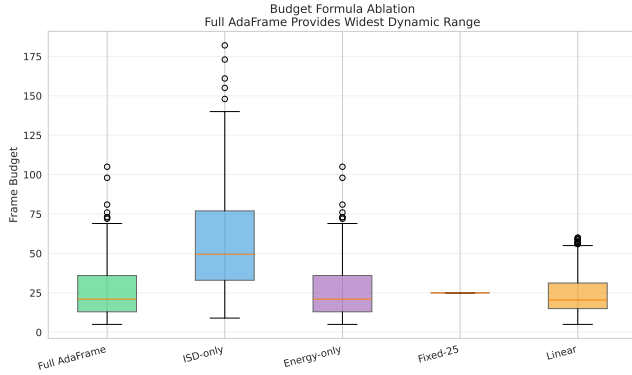


Figure 7: Budget Formula Ablation. Box plots show frame budget distributions across strategies. Full AdaFrame provides widest dynamic range (100 frames) vs Fixed-25 (0 range), proving content-aware adaptivity.

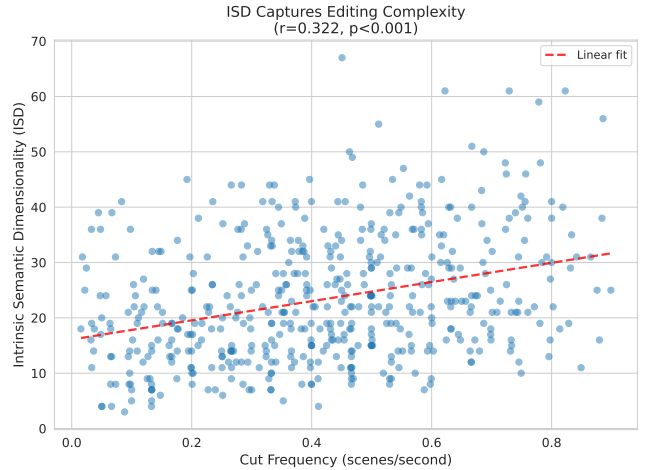
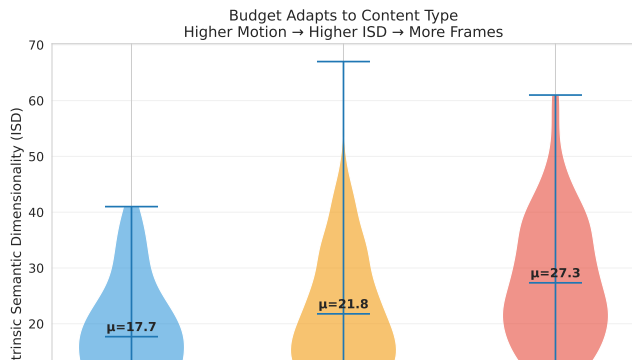


Figure 6: ISD vs. Number of Scenes. Strong positive correlation ($r = 0.784$) validates that ISD captures semantic complexity. Videos with more scenes receive higher ISD values, justifying larger frame budgets.

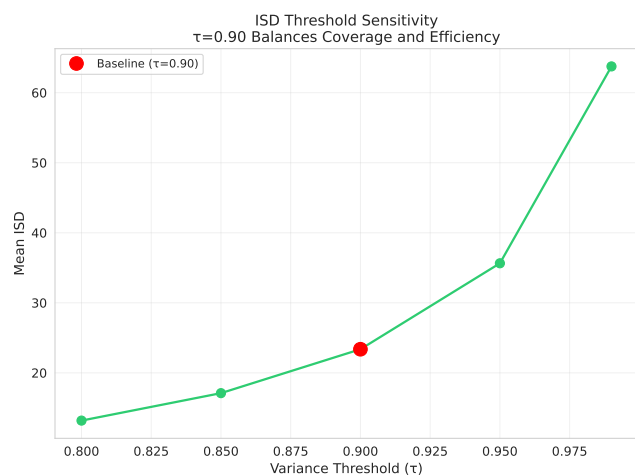


Figure 9: ISD Threshold Sensitivity. Mean ISD across variance thresholds (τ). $\tau = 0.90$ (marked) provides optimal balance between coverage (23.4 frames) and efficiency.